

Columbus Transit & Bike Explorer

An ArcGIS StoryMap and companion web map identifying where Central Ohio's transit-dependent households fall outside walkable access to bus service and safe cycling infrastructure

The Problem

Before this project, transit route planning and demographic need were largely evaluated in isolation from each other. COTA's bus network, Columbus's bike infrastructure, and Census vehicle-ownership data each existed as its own dataset, but nobody had overlaid them to answer the question that actually matters for equity planning: where do households without a personal vehicle also lack walkable access to transit. Without that overlay, pinpointing specific neighborhoods, particularly in east and west Columbus, experiencing compounded mobility isolation meant cross-referencing separate maps and spreadsheets by hand, if it happened at all.

My Approach

I built this as a combination **ArcGIS StoryMap and interactive web map**, so the analysis could be read as a narrative for a general audience while still supporting hands-on exploration for planners who wanted to dig into a specific corridor or neighborhood.

A walkability standard applied consistently, not eyeballed. Every COTA bus stop gets a quarter-mile buffer, the standard five-minute walk distance used in transit planning, generated systematically across the entire stop network rather than assessed stop by stop. That buffer becomes the baseline test for whether a location actually has practical transit access, not just proximity on a map.

Vehicle availability as the equity lens. Census tract and block group data on household vehicle availability is layered directly against the walkability buffers, with tracts where more than 15 to 20 percent of households lack a personal vehicle, well above the roughly 8 percent national average, flagged as high-need zones. Overlapping that threshold with the walkability buffer turns two separate datasets into a single, visual answer to where the gap actually is.

Bike infrastructure treated as part of the access picture, not an afterthought. Columbus's bike network is included and categorized by facility type, since a resident's practical mobility options extend beyond bus stops alone, and the map is built to be read against the city's adopted Bike Plus Plan targets rather than only its current-state network.

A narrative structure for a non-technical audience, with a full explorer underneath. The StoryMap leads a reader through the finding, starting broad with route density and network overview, then narrowing to the specific overlay of transit buffers against vehicle

availability that reveals the actual gap areas. The companion web map underneath gives planners and technical users full layer control to toggle routes, bike facility types, and demographic boundaries on their own.

Technical Highlights

- **Platform:** ArcGIS StoryMaps hosting an embedded ArcGIS Online web map and Instant App explorer
- **Data sources:** COTA GTFS feed for bus stops and routes, City of Columbus Open Data Portal for bike infrastructure, US Census Bureau ACS 5-Year Estimates (Table B08201, household vehicle availability), MORPC Regional Transportation Plan and the Columbus Bike Plus Plan
- **Spatial analysis:** Quarter-mile proximity buffers generated across the full COTA bus stop network, overlaid directly against Census vehicle-availability boundaries to identify compounded access gaps
- **Interactivity:** Toggleable layers for route type, bike facility category, stop buffers, and demographic boundaries, feature pop-ups for routes and Census geography, a StoryMap scroll narrative paired with a full interactive explorer
- **Scope:** Metropolitan Columbus and Central Ohio, covering dozens of COTA routes, thousands of stops, and hundreds of miles of existing and planned bike infrastructure

Why It Matters

An equity gap that isn't visualized against the data that actually defines it, vehicle ownership, walkable transit access, safe cycling routes, tends to stay a matter of general impression rather than something a planner can act on with confidence. Overlaying COTA's stop network buffer against Census vehicle-availability data turned a set of separate, individually unremarkable datasets into a specific, defensible map of where mobility isolation is compounded, work built to directly inform COTA service adjustments, MORPC regional planning, and the city's active transportation investment priorities following adoption of the Bike Plus Plan.